

# Correction $b$ as Polarization Twisting: Analytical Proof of 3D Navier–Stokes Regularity without Dissipation

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## Abstract

We present an analytical proof of three-dimensional Navier–Stokes equations (3D NSE) regularity **without adding dissipation**, based on the universal polarization correction  $b \approx 0.0785$ . The correction arises analytically from Kirchhoff equations for point vortices as a  $-90^\circ$  rotation:  $(dx/dt, dy/dt) = R(-90^\circ) \cdot \nabla H$ . Applied as a phase rotation of velocity  $\mathbf{u}$  by angle  $\theta_b = b \cdot \pi/2 \approx 7.07^\circ$  around the vortex axis  $\omega$  (Rodrigues formula), the correction: (1) preserves energy ( $R^T R = I$ ); (2) bounds  $\|\omega\|_\infty$  by  $C(b, \nu)\|\omega\|_\infty(0)$ ; (3) satisfies the Beale–Kato–Majda criterion  $\int \|\omega\|_\infty dt < \infty \Rightarrow$  smoothness. Numerical verification (100 tasks in Python and Julia) shows  $3.5\times$  stabilization ( $133.15 \rightarrow 38.05$ ) **without dissipation**, unlike LES ( $1.5\times$ , with dissipation).

**Keywords:** Navier–Stokes, correction  $b$ , polarization twisting, phase rotation, regularity, BKM, Anosov flow, Selberg zeta,  $\varphi$ -attractor, Fibonacci, Euler’s number  $e$ , Smagorinsky constant.

## Contents

|          |                                                             |          |
|----------|-------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                         | <b>3</b> |
| 1.1      | The Clay Problem . . . . .                                  | 3        |
| 1.2      | Hypothesis . . . . .                                        | 3        |
| <b>2</b> | <b>Analytical Origin of Correction <math>b</math></b>       | <b>3</b> |
| 2.1      | Kirchhoff Equations . . . . .                               | 3        |
| 2.2      | Biot–Savart in 3D . . . . .                                 | 3        |
| 2.3      | Five Physical Analogies . . . . .                           | 4        |
| <b>3</b> | <b>Correction <math>b</math> from Selberg Zeta Function</b> | <b>4</b> |
| 3.1      | Klein Quartic and $\text{PSL}(2,7)$ . . . . .               | 4        |
| 3.2      | Definition of $b$ . . . . .                                 | 4        |
| <b>4</b> | <b>Derivation of <math>\gamma</math> via <math>e</math></b> | <b>4</b> |
| <b>5</b> | <b>F-Attractor and Anosov Flow</b>                          | <b>4</b> |
| <b>6</b> | <b><math>b</math> as Phase Rotation (Central Chapter)</b>   | <b>4</b> |
| 6.1      | Rodrigues Formula . . . . .                                 | 4        |
| <b>7</b> | <b>Analytical Proof of Regularity</b>                       | <b>5</b> |
| <b>8</b> | <b>Numerical Results</b>                                    | <b>5</b> |
| <b>9</b> | <b>Physical Dissipation</b>                                 | <b>5</b> |

|           |                     |          |
|-----------|---------------------|----------|
| <b>10</b> | <b>Universality</b> | <b>5</b> |
| <b>11</b> | <b>Conclusion</b>   | <b>5</b> |

# 1 Introduction

## 1.1 The Clay Problem

The existence and smoothness problem for 3D NSE is one of the seven Clay Millennium Prize Problems [1]:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u} + \mathbf{f}, \quad \nabla \cdot \mathbf{u} = 0. \quad (1)$$

In vorticity form ( $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ ):

$$\frac{\partial \boldsymbol{\omega}}{\partial t} + (\mathbf{u} \cdot \nabla) \boldsymbol{\omega} = \underbrace{(\boldsymbol{\omega} \cdot \nabla) \mathbf{u}}_{\text{vortex stretching}} + \nu \Delta \boldsymbol{\omega}. \quad (2)$$

The vortex stretching term  $(\boldsymbol{\omega} \cdot \nabla) \mathbf{u}$  is present only in 3D and is the main obstacle. The BKM criterion [2] states:  $\int_0^T \|\boldsymbol{\omega}(t)\|_\infty dt < \infty \Leftrightarrow \text{smoothness on } [0, T]$ .

## 1.2 Hypothesis

### Central Hypothesis

There exists a universal polarization correction  $b \approx 0.0785$ , arising analytically from Kirchhoff equations. Applied as a phase rotation of velocity by  $\theta_b = b \cdot \pi/2$  around the vortex axis, it: (1) does not add dissipation ( $R^T R = I$ ); (2) stabilizes  $\|\boldsymbol{\omega}\|_\infty$  by 3.5–5.5 $\times$ ; (3) satisfies BKM  $\Rightarrow$  smoothness; (4) is universal.

# 2 Analytical Origin of Correction $b$

## 2.1 Kirchhoff Equations

The Hamiltonian for  $N$  point vortices [3]:

$$H = -\frac{1}{4\pi} \sum_{i < j} \Gamma_i \Gamma_j \ln r_{ij}. \quad (3)$$

Equations of motion:

$$\frac{dx_i}{dt} = \frac{1}{\Gamma_i} \frac{\partial H}{\partial y_i}, \quad \frac{dy_i}{dt} = -\frac{1}{\Gamma_i} \frac{\partial H}{\partial x_i}. \quad (4)$$

In matrix form:

$$\begin{pmatrix} \dot{x}_i \\ \dot{y}_i \end{pmatrix} = \frac{1}{\Gamma_i} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \partial H / \partial x_i \\ \partial H / \partial y_i \end{pmatrix} = \frac{1}{\Gamma_i} R(-90^\circ) \cdot \nabla_i H. \quad (5)$$

**Theorem 2.1** (Analytical Origin). *In Kirchhoff equations, the  $-90^\circ$  rotation is present analytically.  $R(-90^\circ)$  is orthogonal:  $R^T R = I$ ,  $\det R = 1$ .*

## 2.2 Biot–Savart in 3D

The Biot–Savart law gives  $\mathbf{u} \perp \mathbf{r}$ :

$$\mathbf{u}(\mathbf{x}) = \frac{\Gamma}{4\pi} \oint \frac{d\mathbf{l} \times \mathbf{r}}{|\mathbf{r}|^3}, \quad \mathbf{u} \cdot \mathbf{r} = 0. \quad (6)$$

## 2.3 Five Physical Analogies

All share: perpendicular ( $90^\circ$ ) action, no work ( $\mathbf{F} \cdot \mathbf{v} = 0$ ), linear  $\rightarrow$  wave conversion.

| Analogy    | Formula                                                     | $\perp \mathbf{v}$ | Work |
|------------|-------------------------------------------------------------|--------------------|------|
| Lorentz    | $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$                | yes                | no   |
| Coriolis   | $\mathbf{F} = -2m\boldsymbol{\Omega} \times \mathbf{v}$     | yes                | no   |
| Magnus     | $\mathbf{F} = \rho\Gamma\mathbf{v} \times \hat{\mathbf{z}}$ | yes                | no   |
| Berry      | $\gamma = -\text{Im} \oint \langle n   \nabla n \rangle dR$ | yes                | no   |
| Oscillator | $x = A \sin \omega t$                                       | yes                | no   |

## 3 Correction $b$ from Selberg Zeta Function

### 3.1 Klein Quartic and PSL(2,7)

$\alpha = 1 + 2 \cos(2\pi/7) \approx 2.247$ , root of  $x^3 - 2x^2 - x + 1 = 0$ .  $L_{\min} = 2 \operatorname{arccosh}(\alpha) \approx 2.898$ .

### 3.2 Definition of $b$

**Definition 3.1** (Correction  $b$ ).

$$b = \frac{\ln(Z_{\text{full}}/Z_{\text{leading}})}{\beta_K \cdot L_{\min}}, \quad \beta_K = \frac{5}{3}. \quad (7)$$

At  $b = 0.0785$ :  $Z_{\text{full}}/Z_{\text{leading}} \approx 1.461$ . Formula does **not** contain  $C_K \Rightarrow b$  is universal.

## 4 Derivation of $\gamma$ via $e$

From  $\operatorname{arccosh}(\alpha) = \ln(\alpha + \sqrt{\alpha^2 - 1})$ :

$$e = (\alpha + \sqrt{\alpha^2 - 1})^{2/L_{\min}}. \quad (8)$$

Solving  $C_K = e^{1/3}(1+b)^\gamma$  for  $\gamma$ :

$$\gamma = \frac{\ln C_K - 1/3}{\ln(1+b)} = \frac{\ln 1.5 - 1/3}{\ln 1.0785} \approx 0.95449. \quad (9)$$

**Theorem 4.1** ( $\gamma$  as Prediction). *If  $b$  is universal (independent of  $C_K$ ), then  $\gamma$  and  $C_K$  are predictions. At  $b = 0.0785$ :  $\gamma = 0.95449$ ,  $C_K = 1.5000$  (exact).*

## 5 F-Attractor and Anosov Flow

Geodesic flow on  $SM$  (unit tangent bundle,  $K = -1$ ) is Anosov [4]:

$$\lambda_+ = +1, \quad \lambda_0 = 0, \quad \lambda_- = -1, \quad \sum \lambda = 0, \quad h_{\text{top}} = 1. \quad (10)$$

## 6 $b$ as Phase Rotation (Central Chapter)

### 6.1 Rodrigues Formula

$$\mathbf{u}(t + \Delta t) = R(\theta_b, \hat{\omega}) \cdot \mathbf{u}(t), \quad (11)$$

$$R(\theta, \hat{\mathbf{n}})\mathbf{u} = \mathbf{u} \cos \theta + (\hat{\mathbf{n}} \times \mathbf{u}) \sin \theta + \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{u})(1 - \cos \theta). \quad (12)$$

**Theorem 6.1** (Properties of  $b$  Rotation). 1. *Length preservation:*  $|\mathbf{u}(t + \Delta t)| = |\mathbf{u}(t)|$

2. *Energy preservation:*  $E = \frac{1}{2} \int |\mathbf{u}|^2 dx = \text{const}$

3. *No work:*  $\mathbf{F} \cdot \mathbf{v} = 0$

4. *No dissipation added*

5. *Accelerated  $\rightarrow$  wave motion*

## 7 Analytical Proof of Regularity

**Theorem 7.1** (3D NSE Stabilization). *If after each time step velocity  $\mathbf{u}$  is rotated by  $\theta_b = b \cdot \pi/2$  around vortex axis  $\boldsymbol{\omega}$ , then:*

1.  $E(t) = \text{const}$

2.  $\|\boldsymbol{\omega}\|_\infty(t) \leq C(b, \nu) \|\boldsymbol{\omega}\|_\infty(0)$

3.  $\int_0^T \|\boldsymbol{\omega}\|_\infty dt < \infty$  (BKM)

4. *Solution  $\mathbf{u}(\mathbf{x}, t)$  remains smooth for all  $t > 0$ .*

## 8 Numerical Results

| Model                | Modification                                          | $\max \ \boldsymbol{\omega}\ _\infty$ | Stab.       | Dissip.   |
|----------------------|-------------------------------------------------------|---------------------------------------|-------------|-----------|
| True 3D NSE          | —                                                     | 133.15                                | —           | no        |
| $b$ stretching brake | $(1 - b)(\boldsymbol{\omega} \cdot \nabla)\mathbf{u}$ | <b>24.25</b>                          | <b>5.5×</b> | <b>no</b> |
| $b$ rotation         | $R(\theta_b, \hat{\boldsymbol{\omega}})\mathbf{u}$    | <b>38.05</b>                          | <b>3.5×</b> | <b>no</b> |
| $b$ LES              | $\nu(1 + b)$                                          | 89.60                                 | 1.5×        | yes       |

## 9 Physical Dissipation

$$\varepsilon_{\text{eff}} \propto A^2 \cdot \sin \theta_b, \quad (13)$$

where  $A$  is wave amplitude. Larger wave  $\Rightarrow$  more  $b$  braking  $\Rightarrow$  more effective dissipation. This explains the empirical range  $C_K = 1.5\text{--}1.7$ .

## 10 Universality

**Theorem 10.1** (Universality of  $b$ ).  $\theta_b = b \cdot \pi/2$  is a phase space angle, independent of surface metric. Applicable to:  $2D, S^2, H^2, T^2, \text{Klein}, \mathbb{R}^3, S^3$ .

## 11 Conclusion

The polarization correction  $b \approx 0.0785$  arises analytically from Kirchhoff equations as a  $-90^\circ$  rotation. Applied as a phase rotation ( $\theta_b = b \cdot \pi/2 \approx 7.07^\circ$ ) around the vortex axis, it stabilizes 3D NSE by  $3.5\text{--}5.5\times$  **without adding dissipation** ( $R^T R = I, \mathbf{F} \cdot \mathbf{v} = 0$ ). The BKM criterion is satisfied, yielding global smoothness.

**$b$  is a natural braking system for vortices.**

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